

Requirements Specification Document (Draft)

Smart Hotel | Apartment Management

System introduction

Smart Hotel cards management software is universal smart card software aimed at management of the hotel and prescribed processes used in hotel management. The management of the hotel from the booking point, guest rooms, rates and payments, etcetera... This solution is intended to simplify management, lower operational costs and improve efficiency. This document seeks to provide guidelines for development of such a system following the key system highlight as explained below: These requirements are evolving/emerging and may change from time to time through highlight of key functional and non-functional requirements. Through engagement with the developer, these specifications may evolve with time and such changes shall be captured along as they occur, during the development process.

Scope

The project covers the hotel management process from making reservations for rooms offline|online, management of payments, ..., automated room locking, data mgt (guests, room, hotel supplies and other logistics), queries for system report generation etcetera...

Required Functionalities

Guests

Guests will be able to check for room's availability(online|offline), view rates, select the rooms, and make payments (or booking). Cancel reservations (as per given terms such as claims for refunds where applicable) . the system will capture guest info and tag it on proximity cards, for access control management ...

Receptionist

Capture, update or modify booking details, assign guest cards.

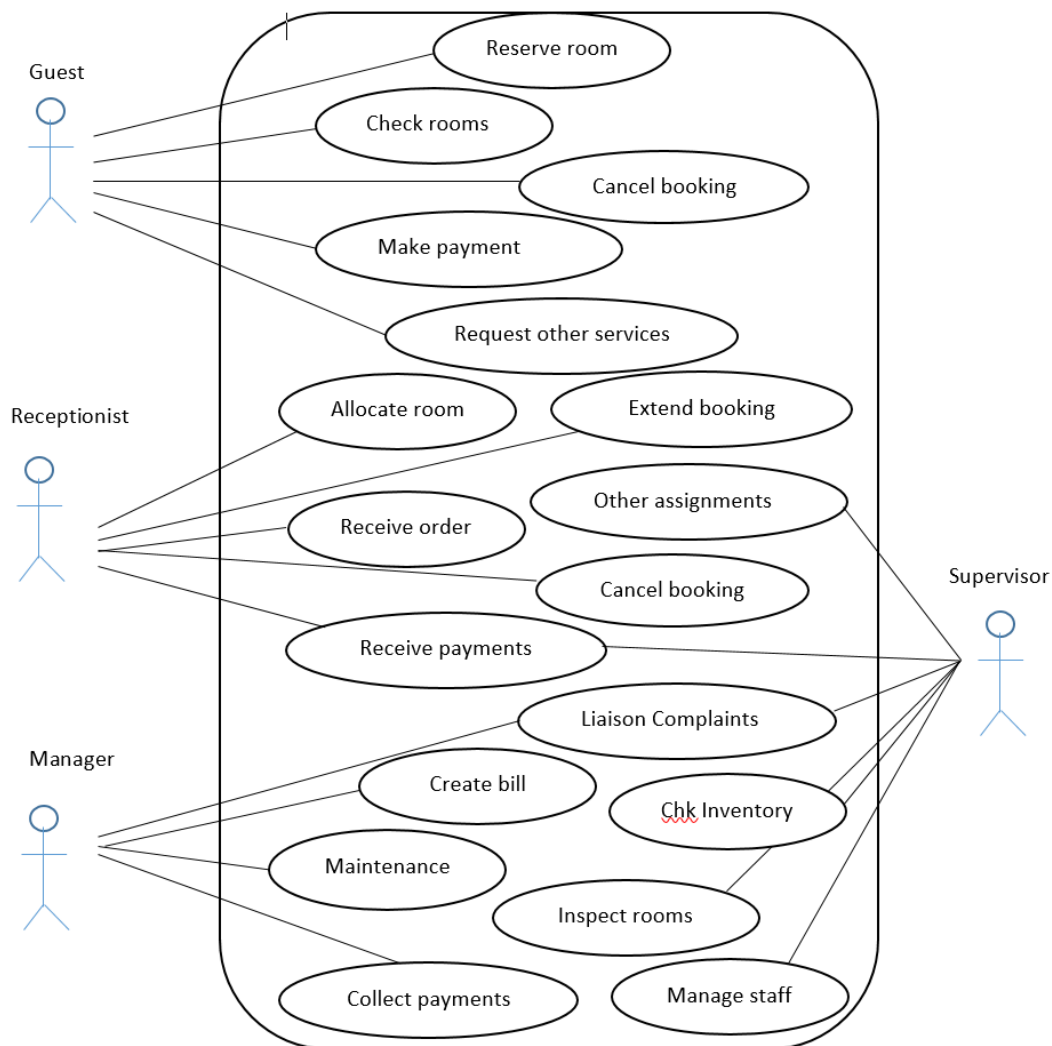
Supervisor

will be able to view the financial report, and be able to update room information such as cost and category. Manage the logistical information, escalate system requests from reception to the manager ...

Administrator|Manager

Provide access privileges on the system, reset passwords and make approvals initiated from system users (supervisor, receptionist ...). Other administrative requirements are explained later in the document. The use case has further details.

Simple use-case



Non Functional Requirements

The system should be capable of replicating the traditional manual hotel management methods. It should be able to perform secure transaction, quick retrieval of information, easy to learn and use, quick to recover from faults| errors and be fault tolerant. It's expected that this lower operational costs and improve the hotel operation efficiency. A few highlights are thus emphasized.

Interface Requirements

System interface requires that the enrollment unit, the system at large and all related intra-sub-systems will satisfy the following requirements:

1. shall be visually appealing
2. shall be quick/responsive
3. shall be consistent (works in the way user expect it to work)
4. be easy to use (navigate)
5. able to provide a 2-way feedback and or sms

The system should put user satisfaction in mind so that all targeted categories of users find it easy to interface with.

Performance Requirements

This defines high level performance needs such as : inter-operability, accuracy, responsiveness in realtime, profile and session handling accuracy.

Security requirements

Security considerations are important while we need to make the system stand out, consequently, evaluations of accuracy of the system in terms of verification of biodata (1:1) and identification (1:N) . these should be swiftly achieved by the developer. Other considerations should take into account the fact underlying the reliability of the system. Apart from the ability to compare biometrics, the legitimacy of data should be emphasized and thus the system shall eliminate possibilities of impersonation during enrollment and minimize passive or active presentation attacks (anti spoofing)

Availability Requirements

The system should be available to all active users (legitimate) at all times and should support and conform to high availability deployment requirements. For example, if the server goes offline, a logged in user may remain functional because of the cached data. The system server too shall be hosted and highly available to process requests.

- Ensure uptime of 99.9 percent to legitimate users
- Ensure a reliable backup and replication system for seamless changeover in case of server failure
- Ensure enough resources are available even at peak hours the system is available and performs to expectations (optimal throughput)

Portability requirements

The developer is assumed to take advantage of different computing platforms and thus use the SDK that supports replication within and across environments and enabling users to port it in different operating environments.

Hosting Environment

- Linux environment or windows environment, but preferably Windows
- Design for Windows 10 or 11 operating systems
- Full hardware support for Android operating environment
- A fingerprint reader that is supported by Free Fingerprint Verification SDK
- Microsoft .NET framework 4.5 (for .NET components)
- Development environment shall be supported by Microsoft Visual Studio .NET, Java SE JDK 8 or better/newer or as may be advised
- Client server environment for easy centralized management
- Database : Mysql or other ...
- Processor : 1.8ghz +
- memory : 1gb +
- screen resolution: 600 pixel 768 pixel(min)
- color 256+

- Installation: via setup.exe wizard (accept terms of license agreement, inst. Path c:\program files with an option to change, any other...)
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Functional Requirements

Enroll users and other information:

The system shall have a biometric device. The system should be able to capture required logistical information, and enroll key users on the system such as guests, receptionists, supervisors and managers among others. Enrolment may be online or offline where necessary and this should be able to do :

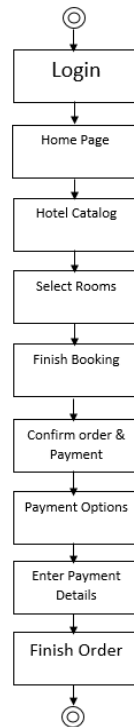
- Facial capture and recognition
- fingerprint capture and verification
- use proximity cards
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The biometric device shall capture the biometric raw data of a user and extract the biometric feature to work with, in this particular case the personal fingerprint and facial biometrics. The different categories of users will be able to use their proximity cards (smart keys) to gain physical access, login to financial accounting systems, use the image processing terminal to gain access and perform any other logging function such as attendance to duty/work

The verification may be by any of the 2 (facial or fingerprint) or as may be otherwise specified by the client.

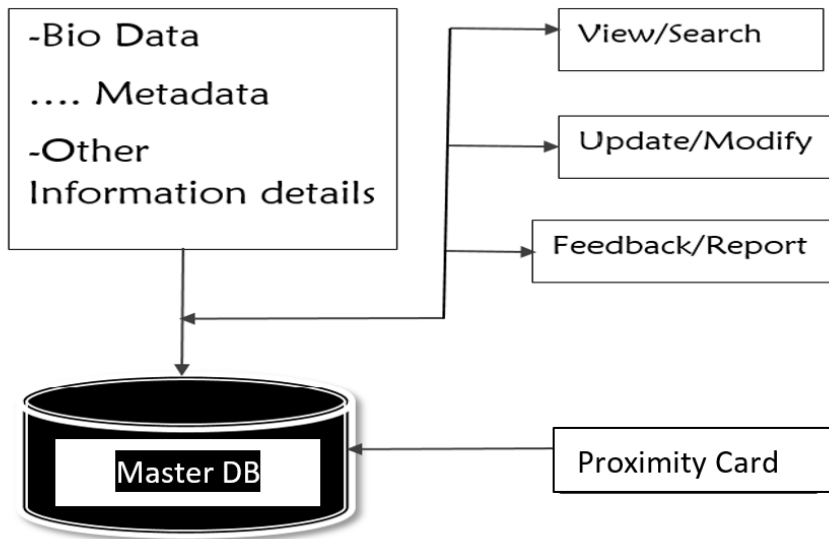
Depending on the ease of data storage, the system may allow enrollment with 1 or 2 fingerprints out of the 10 separately rolled fingers.

Activity diagram



Data storage

Data storage databases will be implemented to capture biometric data, logistical, and other data that facilitates smooth system implementation. The overall architecture shall provide failsafe backup database server mechanism for the independent databases. This is illustrated in the context diagram below:



Other Considerations

Make inquiry; (query) with controlled access (providing multi-angle data inquiry, provide reports from the system)

Perform backup; (path, restore path with datepickers), could be an incremental backup, {auto or manual}

Provide data security; intelligent data management functions {create users via fingerprint or card enrolment, delete, an independent card info database, with card types, room info, checkin status..., user logging, use a default password for account creation (which may later be changed by user)}

Feedback; be able to capture guest feedback and other feedback

Integrated payment gateway; be integrated with **POSSIBLE** local payment gateways such as MoMo platforms, or Visa for wider possibilities. The system shall be able to incorporate a financial component to be able to analyze information captured from payment gateways for financial analyses to help decision making.

Training

The system shall have a training manual. The client and team shall be trained satisfactorily before final handover. Any changes to be made shall be agreed between the developer and client.

Multicurrency Support

The different payment gateways shall provide multicurrency support

Messaging

Notification and messaging thru MMS and email notifications

Reminders

Generate system reminders (payment, checkout etcetera...)

Default Usage

The initial **user** of the system known as "**admin**" with password "admin." Should be created and shall change the password on initial login.

Administrator (with utmost rights –tagged to master control card), super user, user , guest ... administrator can manage rights of other groups

Password strength 6+ characters alphanumeric or as may be advised...

Don't allow copy and paste password

Password recovery options

Provision for system license keys

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Rooms

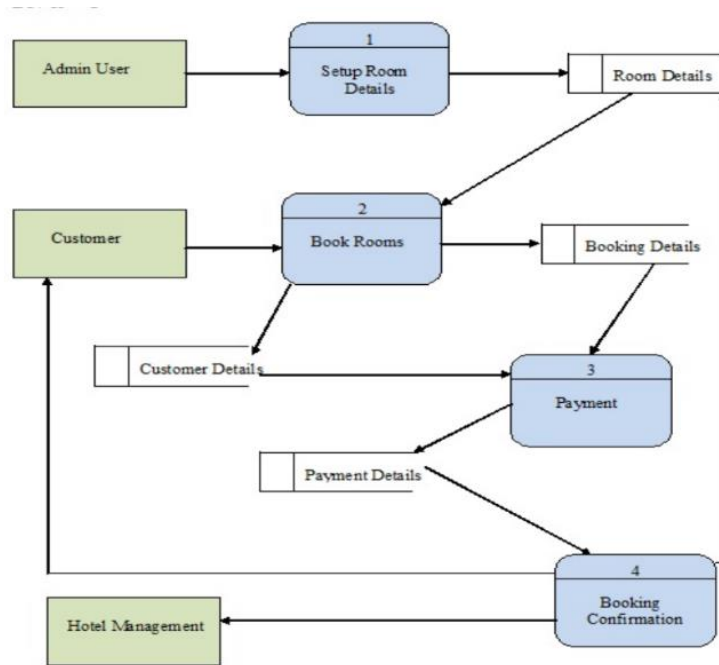
Add room(room type),delete, room type, room status, change room,

Cards

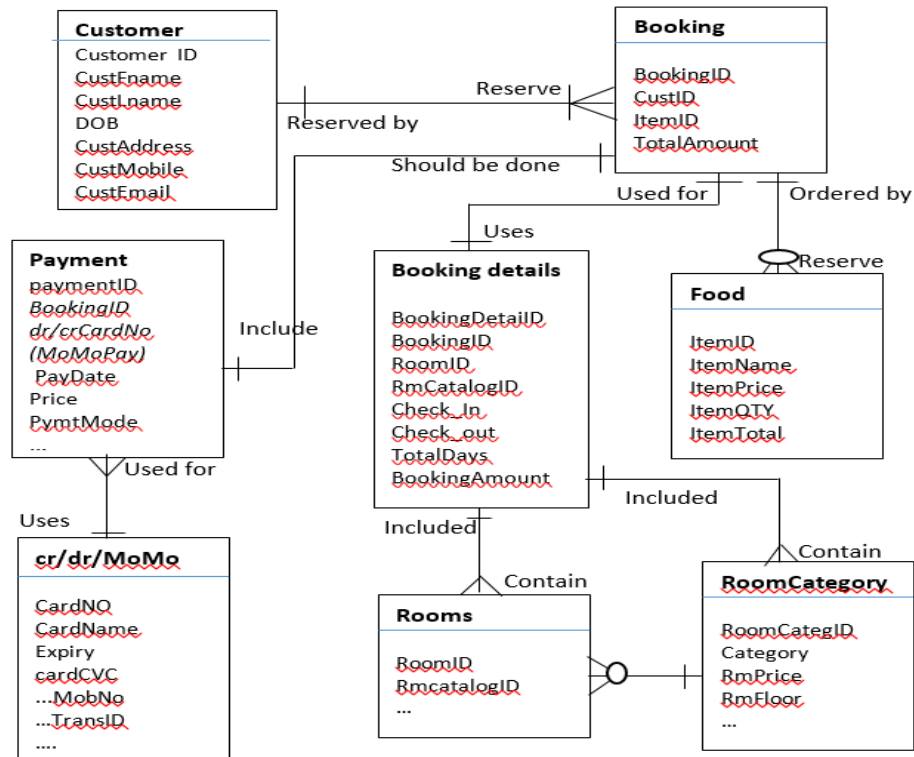
Enroll Add, delete, remove card, card types{admin, super user, user, group, emergency, guest, card status, id(key), search

Error reporting in all system error instances, error logging db

Data flow diagram



ER Diagram for the proposed system



Modules

Module I

Hotel Management: as has been described in this document. This module shall be contracted and development shall not exceed 6 months or **180 days** from the time of engagement.

There shall be a software validation activity to validate the system as it progresses, to measure its conformity with the requirements enlisted.

There shall be a user acceptance test where the developer shall handover the code, user manual detailing the entire physical operation of the system including activated licenses.

The necessary lock hardware should be presented to the developer before commencement for integrity and integration checks. The developer shall also pass-over the associated enhancement code of the sdk and administrative keys to the client, this will be property of the client.

All system changes should be documented as agreed between client and developer

Module II

Apartment Management

Follows similar concept as Hotel mgt module, which is explained. The module shall take care of the few differences:

- Length of stay: the customers for apartments have longer periods of stay
- Rate applicable due to size of apartment offering more living space, such as living rooms, kitchen, furnishing and laundry etc...
- Concierge
- Privacy (hotels are less private): apartments will be more private
- Group stays, for more guests that occupy as a family or company workgroups
- Any other considerations as requirements may change along

This module shall be completed within **30 days** after the completion of the Hotel module. All functional, non functional changes should be documented